Troubleshooting PCA Continuous Monitoring (10107)

- Introduction
- General Points
  - Risk Scores and Risk Levels
  - Linked Jira Issues
  - Tooltips
  - Color Coding
  - Feedback
- Core Reason Codes
  - Attribution Risk
    - Click Spamming
    - Click Stuffing
    - Long CTA
    - Injected Click
  - Event Monitoring IP Risk
    - Data Center
    - IP Reputation
    - Proxy
  - High Risk Login IP
    - Crawler
    - Recent Abuse
    - Proxy
    - Tor/VPN
    - Data Center
    - Bot Activity
  - High Risk Login Email
    - Email Domain Risk
    - Disposable
    - Fraud
  - High Risk URL
    - Low Visit
    - Low Visit Duration
    - Abnormal Bounce Rate
    - Abnormal Traffic Source
    - Malware
    - Copyright Risk
    - Copied Content
    - Low Word Count
  - Performance Geo Mismatch
  - Proportion of Clicks/Actions Outside Listed Media Properties
  - Rejection Rate
  - Abnormal Peaks of Actions/Clicks
  - Repeated IP/UA
  - Paystub History
- Other Tools
  - Account Graph/Fraud Ring

- Appendices

Introduction

The daily PCA Continuous Monitoring job evaluates each active publisher on our platform daily. The output of this job is written to Google BigQuery and is observed via report 10107. It assigns a `final_score` and a `risk_level` to each publisher account on each day, which is derived from a group of underlying risk factors.

This documentation is meant to provide guidance for interpreting the PCA Continuous Monitoring. Using this table and documentation, analysts should be able to properly assess the risks presented by suspicious actors on the platform.

Supplementary risk tools produced by Data Science will be addressed separately.

General Points

Risk Scores and Risk Levels

The risk score is a score from 0-100, indicating the likelihood that the publisher is acting illegitimately. Scores above 20 should be regarded as suspicious, and scores above 40 should be regarded as showing significant signs of likely fraud.

The risk level is an assessment of the overall priority of risk presented by this publisher. The values are “clean,” “mild,” “elevated,” and “critical.” This takes into account the risk score, but also the volume of traffic coming from this publisher. For example, if one publisher has a risk score of 50, but has only sent 100 clicks and 2 actions in the past week, it may only be graded as “mild,” indicating that the publisher’s possible bad actions have relatively little impact and are not as urgent to address. However, a publisher with a risk score of 40 and 10 million clicks in the past week will likely be graded as “critical,” because the threat of the damage they may be doing is greater.

Linked Jira Issues

The Jira issues column will list IRFRAUD issues where this publisher was listed. If you click on the publisher’s name, you will be sent to a list of those issues in JIRA. Please familiarize yourself with these so that you have a sense of the broader context with which we have interacted with this partner. This should also remind you to include all relevant publisher IDs in the data for a given Jira ticket.

Tooltips

If you are unsure of what a column represents, you can consult the tooltip on the column name before referring back to the documentation.

Color Coding

Unfortunately, Operator does not support color-coding columns with drilldowns at this point, so very few columns on 10107 are color-coded. That said, any cell from a column that is colored red-green should be green. If a cell is any other color, that indicates at the very least that further investigation is warranted on that point.

Feedback

Fraud reporting can always be improved. If you have feedback for Data Science regarding either false positives/negatives or additional features you would like to see in reporting, please let us know.

Core Reason Codes

We will group the major risk factors into several groups.

Attribution Risk

Attribution risk refers to cases where we believe that a partner has generated a fake or invalid click on a real end-user's device, hoping that when the end-user organically makes a purchase or other payable action, they will receive a payout, despite not having done anything to help drive the conversion.

It has a number of sub-triggers. In most cases, the telltale signs are a) click-to-action times, and b) conversion rates. The underlying data lives in BigQuery in `ds_coe.attribution_risk_monitoring_new` and can be accessed through this report, which can also be accessed via drilldown in Operator.

Click Spamming

This represents cases where the publisher has an unusually low conversion rate in comparison to the other publishers for the campaign. That means that the clicks are likely not being made intentionally by valid end-users, but are being artificially generated on end-user devices by illicit means, in hopes that some small numbers of the users on whom they are generated will convert. For example, a publisher buying ad space in mobile apps to advertiser installs of other mobile apps may include code within the ads to generate a click event on the ad load, even if

the user does not click the ad. This will lead to an high number of reported clicks, but relatively few actions.

For reference, **a typical conversion rate for a conventional retail campaign will be a few percent and conversion rates well below 1% become suspicious**. However, this will vary from campaign to campaign.

This reason code may overlap with Click Stuffing, because it is looking for cases where a large number of fake clicks are being created.

False Positives: The following are possible sources of false positives:

1. If the publisher is promoting the brand in a radically different way from the brand's other publishers. For example, if a hotel-booking brand is working with a large number of airlines as publishers, and only a small number of content review sites, the high conversion rates of the airlines (which have a captive audience of high-intent consumers who are already committing to travel) may make the other publishers appear to have very low conversion rates. This may especially be the case if the publisher's conversion rate is not very low in the grand scheme of things (e.g., the content site's conversion rate is 1%, while the airlines' are 20%).
2. If the publisher has a large number of internal (e.g., site-health) automated clicks that are not intended to convert, but are being tracked. This is not ideal (we should integrate so that these clicks are not sent to us), but may not be a sign of fraud.

False Negatives: There may be a variety of ways in which fake clicks are masked or otherwise not reported to the platform. Alternately, a brand may have so many illegitimate accounts that it drives their average conversion rates down, making bad actors look normal.

To Debug: Look at conversion rates by relationship. See whether any of the exceptional cases above might apply and whether this conversion rate seems plausible for the relationship.

Click Stuffing

This indicates that we have witnessed, at the event level, signs in tracking that the click was not generated by a normal human user but by a bot or some other automated script. It can mean a variety of things, but typically these are clicks generated through illegitimate means. This will overlap with Click Spamming, because it is an event-level tracking-based attempt to detect bad clicks created at volume rather than an aggregate one.

This term encompasses two different concepts from Event Monitoring, **request_anomaly** and **image_click**. For tracking they are considered separately, but for PCA they are currently combined.

False Positives: There are several non-fraudulent processes which might trigger this reason code:

1. In some cases, the publisher/brand may have an unusual implementation where the actual click event is generated not when the user clicks, but at the end of a waterfall begun by the user. (E.g., a user clicks in-app, the app looks to see if the brand's app is installed, and if not generates a click event leading to the webpage.) These are nonstandard and should be advised against, but are not fraudulent.
2. Similarly, some publishers/brands may have a substantial chunk of size-monitoring bot traffic that could generate bot clicks. These are mostly a problem if the clicks end up getting attribution (which usually doesn't happen) or the publisher has a CPC contract.
3. The publisher or brand may be reporting events to us server-side: i.e., we are not directly tracking the clicks, but are being sent the clicks by a server. This is also not an advised implementation, but it is not necessarily a sign of fraud.

False Negatives: Bad actors, if they are working cleverly, can eliminate the trail of evidence within the click's request properties.

To Debug: Beyond the report linked above, you can further use the columns `reason_code[_count]_request_anomaly` and `reason_code[_count]_image_click` in the brands Event Monitoring to examine this information at the click level. Try to determine whether there is a plausible reason for the relevant relationship to fit into one of the exception cases described above.

Long CTA

Typically, a substantial proportion of users convert shortly after the click event, since the click indicates a moment of elevated interest. Consistently long click-to-action times are suspicious and a sign that the clicks are being generated illegitimately on users' devices in hopes that they will organically convert at some other time. For example, if someone clicks on an ad for a mobile install, they typically install the app within a few minutes; if users are typically taking days or weeks after the last click, that may be a sign that a bad actor is creating fake clicks and hoping that the user will eventually decide (independently) to convert.

False Positives: If the publisher has a promotional method that would lead to abnormally long click-to-action times, relative to other publishers for the campaign, that might be a reason for a false positive.

False Negatives: It is possible that the publisher might configure their activity such that, say, a cookie might erase itself if there isn't a quick conversion.

To Debug: Examine the average click-to-action times against other publishers for the campaign. Make sure that the actual click-to-action time is plausible.

Injected Click

Because the last click is usually what gets credited, bad actors may try to slip in clicks “just in time” prior to a conversion to claim credit. For example, they may use malicious software to determine when someone is on a brand’s checkout page and about to make a purchase, then artificially generate a click. In these cases, we are looking for a combination of fast click-to-action time and overall high conversion rate.

False Positives: False positives tend to emerge in the following cases:

1. The publisher’s implementation is such that conversion can legitimately happen almost automatically. For example, if the user is clicking through to a page where they can authenticate and create an account via Facebook, click-to-action (if the tracked action is account creation) can legitimately occur in seconds, which is unlikely in most situations (e.g., app installs, retail purchases).
2. The publisher is tracking a large number of actions downstream from the original conversion event. This can artificially increase the conversion rate.
3. The publisher is promoting in a way that is very different from other publishers for the brand, such that they would have a high conversion rate and low click-to-action times.

False Negatives: Because our system has two conditions, it is possible that click injection could occur without tripping both triggers.

1. For instance, the click might be injected a minute or two pre-conversion, which would still likely get credited but might not appear obviously suspicious.
2. The conversion rate might be higher than average, but not so abnormally high for the campaign that it would be flagged. This is especially true for cases where the campaign in generally has a high conversion rate.

To Debug: Examine average click-to-action time and conversion rate by relationship against other publishers for the campaign to determine whether there is a plausible explanation for the publisher’s numbers.

Event Monitoring IP Risk

Some bad actors may generate click and transaction activity via proxies, servers, or compromised IPs. They may do this for several reasons, some of which might include:

1. They are generating fake activity via scripts hosted on various machines, for example to inflate their numbers on CPC contracts

2. They have control of a botnet of compromised end-user devices to distribute illegitimate actions across multiple locations to make it seem more realistic
3. They are trying to hide their own location, especially in the case of fake leads or credit card fraud where they are creating fake actions

The underlying data can be found in the Singlestore `agg` tables and this report.

Data Center

The IP belongs to an ISP on our in-house (Forensiq) data center list, such as Amazon Web Services. Typically, normal users should not be accessing the Internet via data center IP ranges. Large numbers of clicks coming from data centers indicate that the clicks and actions being generated are very unlikely to have been generated by valid end users.

False Positives: A number of accounts on Impact have server-side integrations, which means that the clicks may be directed to data centers or otherwise recorded internally before being sent to us. This is not supported, but it may not be a sign of fraud per se. Some sites may also be sending us, intentionally or not, clicks deriving from high-volume internal site tests or external scrapers. Furthermore, it is entirely normal for a small portion of a site's clicks (1-2% or so) to come from a data center, as many sites are frequently being indexed or crawled by a variety of user scripts.

False Negatives: Our data center list is manually updated, and there may well be data centers that it does not include. If you find such data centers, please report them to Data Science so they can be included in a future update.

IP Reputation

This comes from a proprietary internal algorithm, which essentially tracks whether there is information from JavaScript indicating that the IPs users come from all around the world. This is a sign that the IP has been compromised or is being used as a open proxy. This should not represent a large percentage of user traffic for a given publisher. If it does, that indicates that the traffic may be created by a botnet or bad actors concealing their identities.

False Positives: There are some regions of the world where IP behavior functions in a manner sufficiently different from typical use that normal use may make the IP look suspicious. (For example, at one point we frequently flagged Asian traffic because its users were using default California computer settings.) If the traffic is geographically unusual, that may create a false positive.

False Negatives: Currently, the algorithm is only as strong as its input data, which derives from legacy Forensiq clients. This is a small dataset, and much Impact traffic may not be covered by

it.

Proxy

This reason code derives from scraped public lists indicating that the IPs either belong to known proxy networks or have been reported as generating Internet spam. If these represent a significant part of a publisher's traffic, this is likely to indicate that the publisher runs various spamming or otherwise illegitimate web operations.

False Positives: These should be rare.

False Negatives: These are old lists and collection methods, and likely are not particularly comprehensive.

High Risk Login IP

Bad actors will frequently seek to mask their identity (e.g., their geographic location) by using proxies, servers, or compromised IPs. Consequently, we score recent publisher login IPs against several services to determine whether the IPs used by publisher are suspicious.

The underlying data can be found in BQ in `ds_coe.login_ip_risk_monitoring` and in this report, which can be accessed via a drilldown.

Crawler

Either our internal (Forensiq) servers or IPQualityScore's have identified this IP as belonging to a corporate crawler (e.g., Google Search). These should not, under any circumstances, be the source of a login from an account.

False Positives: This should only very rarely be flagged for logins. If it is, then it is possible something odd is happening and we should take a closer look.

False Negatives: The IP may have been sold to a new ISP since we updated our internal lookup database.

Recent Abuse

This means that IPQualityScore has detected abusive behavior from this IP in its various traps and honey pots. This likely means that the IP is sending out automated malicious traffic of various kinds all over the Internet, since the honey pots are not intended to attract valid users.

False Positives: This is very unlikely to be a false positive. If it is, it means that IPQualityScore has simply made an error, which we unfortunately have little insight into.

False Negatives: IPQualityScore may simply not have seen traffic from the IP.

Proxy

Some combination of IPQualityScore, Scamalytics, and our internal system (Forensiq) has flagged the IP as an open proxy. This means that it is vulnerable to other IPs funneling traffic through it, and bad actors might use it to mask their activity.

False Positives: These may occur if the user is logging in with a corporate VPN, which many of the services will treat as a proxy IP, or is using a VPN to escape Internet controls in a country like China.

False Negatives: Our flagging is limited to the IP databases of these vendors and requires confirmation by at least one other. Gaps in these databases will lead to gaps in monitoring.

To Debug: Determine whether the publisher's use case might fit into one of the exceptions described above. In general, we advise that (if they have the option) users do not use a VPN to sign onto our platform.

Tor/VPN

IPQualityScore and/or Scamlytics have reported this IP as belonging to a known VPN or as a TOR exit node. These typically mean that the user is attempting to mask their identity or location. Often, bad actors will do this to hide their true identities (e.g., a Nigerian account impersonating an American).

False Positives: See above

False Negatives: See above

To Debug: See above

Data Center

The IP belongs to an ISP on our in-house (Forensiq) data center list, such as Amazon Web Services. Typically, normal users should not be accessing the Internet via data center IP ranges. If someone is logging on with a data center, it is probably to mask their identity.

False Positives: See above

False Negatives: See above

To Debug: See above

Bot Activity

Some combination of IPQualityScore and our internal system (Forensiq) has flagged this IP as being the source of historical malicious traffic. This might be a sign that either this is the IP of a malicious actor or the IP of an end-user compromised by malware.

False Positives: Sometimes, regular VPNs (due to their high volume of traffic) are flagged under this reason code. See above for further discussion.

False Negatives: See above

To Debug: See above

High Risk Login Email

This means the user's email address was flagged by IPQualityScore or our internal blocklist. Generally, we will want to see email addresses from either major free providers or established corporate addresses, which have a history of stable and non-malicious activity. Frequently, bad actors will either:

1. Adopt burner emails to mask their identity, cycling through them as necessary for different illegitimate activities.
2. Send a large amount of malicious activity (e.g., email spamming) from an email, indicating that it does not belong to a legitimate actor.

The underlying data can be found in BQ in `ds_coe.login_email_risk_monitoring` and via this report, which is available as a drilldown.

Email Domain Risk

This indicates that the email's domain is on our in-house blocklist. The blocklist has been assembled by Compliance over time as it has periodically noticed patterns of bad accounts applying with similar email addresses.

False Positives: This list may sometimes include free email domains that are commonly abused by fraudsters, but which also have legitimate users. (Historically, it has included outlook and hotmail, both of which had to be removed).

False Negatives: Since this list is reactive and manually maintained, it will naturally lag fraud efforts.

To Debug: Try to find the source domain and determine whether any normal valid users might sign up using that domain.

Disposable

This indicates that IPQualityScore has evaluated the email as "disposable," meaning it is an email likely made through a service that allows the generation of a large number of unique email addresses quickly. Bad actors will often use these as burner addresses to avoid having their bad

activities linked together. If we see a disposable address, we should not trust the person using it.

False Positives: Since we don't have access to the detailed methodology that IPQ are using, it's possible that they could flag users whose accounts are relatively new or with odd email handles or email domains. This is unlikely.

False Negatives: IPQ won't have a complete database of all disposable emails.

To Debug: Try to find the source domain and determine whether any normal valid users might sign up using that domain.

Fraud

This is indicates that one or more fraud triggers from IPQualityScore have been tripped. These might include emails with a history of sending spam emails, DNS resolution that does not check out, etc. If any of these tests flags the email, we should be suspicious of whoever is using it.

False Positives: It's likely that these flags are generated by looking at mass user data and detection from abnormal usage patterns. Since we don't have access to the detailed methodology that IPQ is using, it's possible that they could have an incomplete or erroneous picture

It's also possible that certain corporate accounts may be linked to certain bot-like activities online, such as sending large volumes of email with very low open rates.

False Negatives: IPQ won't have a complete database of all fraudulent emails.

To Debug: Although we won't have access to IPQ's database, you might examine the domain the email comes from to determine if it is an obvious false positive or looks inherently suspicious. For example, an email owned by a trusted business is probably fine, but an email whose domain looks unfamiliar, hastily assembled, or linked to a high-risk geographic region is more likely to be a true positive.

High Risk URL

Bad actors may use illegitimate web domains to plant fake clicks on unsuspecting users who have been redirected away from their intended website by malware or other illicit means. In these cases, if there are signs that the referring websites are suspicious (i.e., unlikely to have a base of valid user traffic), we will flag the traffic. To determine whether that is the case, we source aggregate data estimates of the website traffic from:

1. SEMRush to determine if there is valid traffic on the site.
2. Netskope to categorize potential security risks.

3. Copyscape to determine whether the site has been largely plagiarized and set up as a front from another website.
4. The MalwareTips blog to determine if the site is used primarily within scam operations.
5. Google's list of strikedown requests for websites publishing copyrighted materials.

The underlying data can be found in BQ in the table `ds_coe.url_risk_monitoring`, and in this report , which is accessible via drilldown. Right now, the report somewhat awkwardly overlaps with the Domain Reputation reason code in Event Monitoring. Currently, we will separately present the results of Domain Reputation evaluating the clicks and url_risk evaluating actions.

Low Visit

This indicates that the website does not, according to SEMRush, have much traffic. If that is the case, it is unlikely that it is sending us referrals, which means that the traffic deriving from here is likely to be unnatural in some respect.

False Positives: SEMRush's data is imperfect, especially for smaller sites. They may be underestimating the amount of traffic for the site.

False Negatives: These will be rare. If there is little traffic for the site, SEMRush is unlikely to overestimate it.

To Debug: Look at how recently we have queried the data, and try to cross-check against a publicly-available alternate provider (e.g., the free version of Similarweb), or against any other source we may have, to validate whether we think this site has actual traffic.

Low Visit Duration

This indicates that the average visitor is only on the site for a few seconds at a time. This typically indicates that the source is largely bots or users who were redirected against their intentions. Either way, these visitors are unlikely to be legitimately using the site to get referrals for products and services.

False Positives: SEMRush's data is imperfect, especially for smaller sites. They may have erratic data for time on site that changes month to month.

False Negatives: Same as the above

To Debug: Look at how recently we have queried the data, and try to cross-check against a publicly-available alternate provider (e.g., the free version of Similarweb), or against any other source we may have, to validate whether we think this site has actual traffic.

Abnormal Bounce Rate

A normal website likely has some users who are browsing multiple pages, and some user who just go to one. If a very large percentage of users are either bouncing immediately or are rarely ever bouncing, this is likely a sign that the website's traffic is mostly bots following the same path. We should not trust this traffic.

False Positives: See above.

False Negative: See above.

To Debug: See above.

Abnormal Traffic Source

A normal website with engaged users will typically have incoming traffic from a variety of sources. Some will likely navigate directly, others will search, others will click on paid ads, etc. If almost all of the traffic is navigating to the site via a single method (especially if it's direct), that may be a sign that the traffic is mostly bots.

False Positives: See above.

False Negative: See above.

To Debug: See above.

Malware

If we are seeing a referrer URL that has been reported elsewhere (either by MalwareTips or Netskope) as a site that is used as part of a scam, adware, or some other security risk, it is likely that any traffic we see referred from there has been generated illegitimately.

False Positives: These should be very rare, and would only occur as a result of an incorrect judgment from Netskope or from Data Science in adding a site to a blocklist.

False Negatives: Netskope data is imperfect. DS only adding sites that our scans have picked up to add manually to the malware list. Either of us may have missed something.

To Debug: Search for references to the site online and see how the site has been categorized by others. You should be able to corroborate fairly easily whether or not the site itself appears to be malicious and whether it should appear on our list.

Copyright Risk

Websites that host large amounts of pirated material or other contents to which they do not hold copyright will often try to monetize through questionable means. If either Google's strikedown

request list or Netskope has identified that the site is primarily used for hosting this kind of material, in general, we do not want to do business with it.

False Positives: It is possible that Netskope has miscategorized the site, or it is also possible that it has a significant minority of user-generated content (some of which may be uploaded illegally by users) that is not indicative of the site as a whole.

False Negatives: It is possible that the number of URLs reported to Google is not sufficient for us to have flagged the domain, or that Netskope has not successfully found the security risk posed by the site.

To Debug: It should be pretty ease to tell from a quick visit to the site whether or not it is primarily a repository of pirated content.

Copied Content

There are two reasons why a bad actor might copy another website. One is that by copying a moderately successful site, they may be able to siphon some of its search traffic without having to spend effort on creating their own content. Another reason is that it is intended to be a front to mask illegitimate traffic generated elsewhere. If Copyscape has found one or more webpages across the web that this domains appears to have copied its content from, we should be suspicious of the site.

False Positives: These are not uncommon. Copyscape is often mistaken about which site has copied the other, leading legitimate sites to be accused of copying the illegitimate sites that in fact copied them. Also, legitimate webpages may be generated from a common template, which can generate false positives.

False Negatives: It is possible that some, but not all, content may be copied, which might end up showing as benign to Copyscape.

To Debug: Take a look at the flagged site and the site it is alleged to have copied from. It should be obvious whether the plagiarism charge is legitimate.

Low Word Count

If a website has minimal text, that may be a sign that it is a hastily-assembled front for illegitimate activity. As part of the Copyscape process, we also record the number of words reported on the homepage, and if this falls below a threshold, the site is flagged.

False Positives: These are somewhat common, unfortunately. Reasons that a site may be legitimate but reported as having minimal content:

1. The homepage does not have a lot of text, but subpages do.

2. The homepage includes its text within images rather than a typical HTML text field.
3. The site is legitimately made from a low-text template.
4. Copyscape has not successfully scraped the page's text.

False Negatives: The site could have enough boilerplate text to escape detection.

To Debug: Take a look at the flagged site and see if it seems to have any actual content. This should be fairly obvious from a quick glance.

Performance Geo Mismatch

If the geographic region of the IP of the converting action is not among the geographic regions listed among either the publisher's or brand's promo countries, that may be a sign that:

1. The publisher has misrepresented their location to appear more desirable to the brand.
2. The publisher is claiming to promote in a region relevant to the brand, but is generating fake orders or clicks in their own region.
3. The publisher is maliciously generating bad traffic from available IPs around the world rather than appealing to the brand's target audience.

The underlying data can be found in BQ in `ds_coe.geo_mismatch_monitoring` and in this report, which is accessible via drilldown.

False Positives: These are somewhat common. Many publishers and brands have not exhaustively maintained their list of promo countries, and many brands do not especially care whether they do or not. As such, this code has relatively little weight in the algorithm.

False Negatives: We only look at region-level discrepancies. If a brand has strong views about promotion within a region (e.g., Germany vs. Spain), this algorithm will not flag them.

To Debug: Look at the region the actions are coming from, as against the purported audience being appealed to by the brand and publisher. Think about whether there is a plausible reason for a mismatch (e.g., traffic to a brand based in Spain from a Spanish-speaking region elsewhere in the world) or not.

Proportion of Clicks/Actions Outside Listed Media Properties

When the referring URL of the credited click comes from a domain that is not listed among the media properties of the publisher, it may be a sign that the publisher is a bad actor. If the publisher's listed media properties are merely fronts without an actual user base, then the traffic probably originates elsewhere, whether from malicious sites or malicious techniques used to draw traffic from legitimate sources (e.g., ad-hijacking a search engine).

The underlying data can be found in BQ in `ds_coe.media_property_risk_monitoring`, and in this report, which is available via drilldown.

False Positives: These are common, as many publishers do not exhaustively list all their media properties. Alternately, our referral URL may come from a known and legitimate intermediary tracking page rather than the page of the original click.

False Negatives: Often, we are not able to record a referral URL (especially from an app environment), which limits our insight into this issue. Furthermore, sophisticated publishers may funnel traffic generated in illegitimate places through their own tracking domains, which they are free to list.

To Debug: Look at the reported referral URLs (and possibly also landing page URLs) and see if there is a plausible explanation for the traffic being funneled through that source instead of the listed media properties. Determine who owns the referral URLs, if possible, and try to work out what the intended traffic path is for this relationship.

Rejection Rate

If a substantial percentage of the publisher's actions have been reversed (i.e., the brand(s) have reversed credit for the action prior to the locking window's closure), that may mean the publisher is illegitimate. There are several possible reasons for a large-scale reversal of actions:

1. One of more brands may have discovered something objectionable about the publisher and decided to promptly reverse all actions.
2. The publisher may be engaging in credit card fraud, purchasing items for free and subsequently getting a kickback on expenses they have made themselves.
3. The publisher may have purchased the items in hopes of returning them for refunds while keeping the commissions.

The underlying data can be found in BQ in `ds_coe.rejection_rate_monitoring`, and in this report, which is available via drilldown.

False Positives: There may be daily fluctuations in traffic, especially for low-volume partners, in which a rejection rate might spike briefly. For example, if there happens to be two unrelated, innocent reversals on a day in which a partner has only ten purchases, this might give the publisher a high reversal rate. Also, since the reversals originate from a brand, it is possible that the brand has made a mistake in reversing the publisher's commissions.

False Negatives: This reason code is reactive: it only notices a problem after one brand has reacted. This means it will not notice things like credit card fraud until the horse is out of the

barn. It is mostly useful to help escalate response to other publishers.

To Debug: Try to determine the specific reason for the reversals. Reach out to CSMs if there is not sufficient data in the underlying tables.

Abnormal Peaks of Actions/Clicks

This trigger refers to cases where the publisher's traffic has spiked recently. This might indicate a surge of illicit activities outside of their normal or nominal promotional activity. If a publisher suddenly becomes a top performer without a prior history, or if the publisher sees a spike without a corresponding spike in activity for the rest of the program, that often means the traffic is not real.

The underlying data can be found in BQ in `ds_coe.abnormal_peaks_monitoring`, and in this report, which is available via drilldown.

False Positives: There are a number of legitimate reasons why a publisher's activity might spike (a new promotion, ramping up their business, adding new campaigns).

False Negatives: These should not occur, obviously, since we have the full ability to track activity volumes on the platform.

To Debug: First, look at the publisher's recent activity for the most-affected campaigns, then contact the relevant account managers to see if they are aware of a reason for the spike.

Repeated IP/UA

If most of a publisher's traffic is coming from a small number of IPs or device types, that may indicate illegitimate activity. Most publishers should attract a variety of different users. If we are seeing repeated IPs or user-agents, that could mean:

1. The publisher is making all the orders themselves. This might indicate stolen credit cards or money laundering, among other things.
2. The traffic is largely coming from a set of servers or some other kind of click farm.

This signal produces a risk score from 0-30, with 30 being the most heavily concentrated in a small number of IPs and user-agents.

You can consult this report for details, which is available via drilldown.

False Positives: There are a certain number of reasons why we might see this kind of behavior. First, with small partners, a small number of orders from the same user may not be suspicious. Even from larger partners, they may have a certain number of power-users who repeatedly

purchase from the brand. Second, the bad actor may be an end-user who is not related to the publisher. This will typically be in the case when we see this flag only for a short period of time.

False Negatives: The standard for how many orders is too many will be different from brand to brand. Some brands may expect some repeated purchases, while others will expect IPs to be consistently unique. What we consider to be too many purchases may not be the same as the brand.

To Debug: Take a look at the events in question and determine whether they appear to be plausible behavior for this relationship.

Paystub History

This is a less-common trigger, indicating that Paypal has been repeatedly rejecting payments to the publisher for reasons that indicate potential fraud.

You may consult the underlying data in this report, which is available via drilldown.

False Positives: These should be rare. It might indicate a false positive on Paypal's part, or that the publisher has not updated their financial information recently.

False Negatives: Since we require that Paypal has already flagged the account, this will obviously not flag a large number of potentially fraudulent publishers who have not been flagged by Paypal, or indeed do not use Paypal.

To Debug: Check the paystub history in the report and try to determine if there is a legitimate reason for the Paypal payment to have been refused.

Other Tools

Account Graph/Fraud Ring

This looks for cases in which many accounts are linked to the same actor (i.e., linked by email/IP/financial info), despite claiming not to be linked. This is often a sign of someone creating front accounts for potential financial crimes or otherwise trying to create partnerships illegitimately by masking their true identity. The report can be found here.

False Positives: We make an effort to limit the final score for cases where one parent company or agency is controlling many accounts legitimately, but it may be the case that the account details (name/address/parent/branding/agency ID) do not clearly link these accounts. These cases will be false positives.

False Negatives: This algorithm depends upon the bad actor not changing their IP/email/financial info from account to account, or otherwise giving the accounts similar names.

If they are careful to do this, we will not link them.

To Debug: In the report linked above, isolate the graph that includes this account and see a) what other accounts link to this one and via what common characteristics, and b) if there is a plausible reason for this result (e.g., they all share the same agency or parent company).

Appendices

Here is other documentation related to assessing risk factors:

1. Many of elements of PCA begin their lives as elements of the Forensiq risk product. The false-positive/negative documentation for that lies here:
   a. False Positive Reports
   b. False Negative Reports